

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Bhaskara Contest

(Maths Olympiad at Junior Level for High School Classes)

Saturday, August 31, 2002.

e:

- 1) Fill in the answer sheet with your Name, Class, Section etc. in the specified places.
- 2) Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- 3) For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.
- 4) Diagrams are only visual aids; they are not drawn to scale.
- 5) You are free to do rough work on separate sheets.
- 6) Duration of the test: 2 p.m. to 4. p.m.

If the graph of $y = m^x$ passes through the points $(2,3)$ and $(3,n)$ then mn equals to

- A) 3 B) 6 C) 9 D) 8

When the integers from 1 to 1000 are written on paper, the number of zeros used in writing is

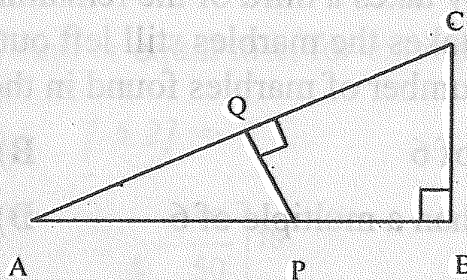
- A) 200 B) 144 C) 192 D) 500

Exactly five interior angles of a convex polygon are obtuse. The maximum possible number of sides for such a polygon is

- A) 8 B) 10 C) 6 D) 9

In the figure $\angle Q$ and $\angle B$ are right angles. If $AQ=15$, $BC=16$, $AP=17$ then QC is

- A) 18 B) 16
C) 19 D) 12

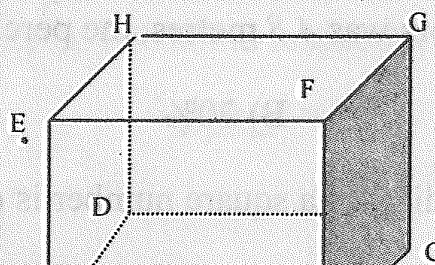


In a right angled triangle, if the square of the hypotenuse is twice the product of the other two sides, then one of the angles of the triangle is

- A) 30° B) 45° C) 60° D) 15°

In the cube shown in the figure, if the length of diagonal AG is $\sqrt{12}$, the volume of the cube is

- A) $8\sqrt{2}$ B) 8



7. The number of ten digit numbers, whose sum of the digits is 2, is
 A) 2 B) 8 C) 12 D) 10
8. The value of $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{99 \times 100}$ is
 A) less than $\frac{99}{100}$ B) equal to $\frac{99}{100}$ C) greater than $\frac{100}{99}$ D) equal to $\frac{100}{99}$
9. Given that $a+b=1$ (where a, b are real numbers), the maximum value of ab is
 A) 1 B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) $\frac{3}{4}$
10. $\frac{a^{-4} - b^{-4}}{a^{-2} - b^{-2}}$ is same as
 A) $\frac{b^2 - a^2}{a^2 b^2}$ B) $\frac{b^2 + a^2}{a^2 b^2}$ C) $\frac{a^2 b^2}{a^2 + b^2}$ D) $\frac{a^2 b^2}{b^2 - a^2}$
11. Three boys agree to divide a bag of marbles as follows:
 The first boy taken one more than half of the marbles;
 The second boy takes a third of the remaining marbles;
 The third boy takes the marbles still left out in the bag.
 The original number of marbles found in the bag in the beginning should have been
 A) a multiple of 6 B) one more than a multiple of 6
 C) two more than a multiple of 6 D) three more than a multiple of 6
12. The sum of the digits of the number $1000^{20} - 20$, expressed in decimal notation is
 A) 530 B) 520 C) 510 D) 500
13. In a long jump contest, the average jump length of the successful competitors was 6.4 metres and that of the unsuccessful competitors was 4.4 metres. If the average jump length of all the competitors was 4.8 metres, the percentage of successful competitors is
 A) 25% B) 20% C) 16% D) 4%
14. The unit digit of a square number is 6. Then the tens digit of the number is an

The units digit of the product of all the prime numbers lying between 1 and 1000 is

- A) 0 B) 2 C) 3 D) 6

If a sector of a circle rotates around the centre of the circle in movements of 75° , the number of sector movements needed for the sector to come back to its original position is

- A) 16 B) 20 C) 24 D) 36

The measures of three consecutive angles of a quadrilateral are as 2:3:4. the fourth angle θ is

- A) $0 < \theta < 90^\circ$ B) 90° C) $90^\circ < \theta < 180^\circ$ D) cannot be determined

Four digit numbers are formed using the digits 1,2,3,4 (repetitions among these digits allowed). The number of such four digit numbers divisible by 11 is

- A) 8 B) 12 C) 28 D) 36

Given that $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

Then the quadratic equation $x^2 + 2|x| + 1 = 0$ has

- A) one positive and one negative roots B) both roots positive
C) both roots negative D) no real roots

The symbol $[x]$ denotes the largest integer less than or equal to x , for example,

$$[5] = 5, \quad [4.7] = 4, \quad [-5.2] = -6$$

Then, the value of $\left[\frac{1}{4}\right] + \left[\frac{1}{4} + \frac{1}{50}\right] + \left[\frac{1}{4} + \frac{2}{50}\right] + \dots + \left[\frac{1}{4} + \frac{40}{50}\right]$ is

- A) 40 B) 10 C) 3 D) 0

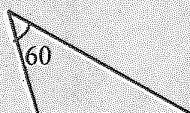
The quotient and remainder when $x^{2002} - 2001$ is divided by x^{91} are, respectively

- A) x^{91} , -2001 B) $x^{91 \times 22}$, 2001 C) $x^{91 \times 21}$, -2001 D) x^{91} , 2001

For $x^2 + 2x + 5$ to be a factor of $x^4 + px^2 + q$, the values of p and q should respectively be

- A) 2,5 B) 5,25 C) 6,25 D) 5,2

In the given figure, $m\angle C$ is



24. A and B had a pack of playing cards. A gave B some cards, took some himself and then laid the rest of the pack on the floor. B had more cards than A . If A gave a certain number of his cards to B , then B would have four times as many cards as A . But if instead B gave the same number to A then B would be left with three times as many cards as A . How many cards did B give to A in the beginning?
- A) 31 B) 9 C) 26 D) 13
25. In a school of student strength 500, two-thirds of students who do not wear spectacles do not bring lunch; three-quarters of the students who do not bring lunch do not wear spectacles. Altogether 60 spectacted students bring lunch too. The number of students who do not wear spectacles and do not bring lunch is
- A) 240 B) 120 C) 360 D) 90
26. Men, women and children numbering one hundred in all were distributed Rs. 1000. Each man got Rs. 50, each women got Rs. 10 and each child got Rs. 0.50. How many children were there?
- A) 50 B) 80 C) 1 D) 19
27. Cost of $\frac{1}{4}$ box of sweets is the same as that of 6 pencils. Mr. Wise purchased $\frac{1}{4}$ box of sweets and 5 pencils. He found that the total of the digits of the amount he has to pay is 8. He gave Rs. 50 to the shop-owner and got the balance. What is the cost of a pencil?
- A) 4 B) 8 C) 6 D) 5
28. P and Q are school students. If one subtracts Q 's age from the square of P 's age the result is 158 but if P 's age is subtracted from the square of Q 's age the result is 108. The age of P is
- A) 11 B) 12 C) 13 D) 14
29. $P(x)$ is a polynomial satisfying $P\left(x + \frac{3}{2}\right) = P(x)$, for all real values of x . If $P(5) = 2002$, then $P(8)$ equals
- A) 2002 B) $2003\frac{1}{2}$ C) $\frac{2003}{2}$ D) $2002\frac{1}{2}$
30. A convex polygon has 44 diagonals. The number of its sides is
- A) 10 B) 11 C) 12 D) 13